

**Sensemaking during a law enforcement officer's use-of-force: A qualitative interpretive,
thematic computational analysis on officer involved-shootings in California using AB1506
reports**

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Abstract / Summary

After the murder of George Floyd by a Minneapolis police officer and mounting pressure for greater accountability, the California State Legislature passed AB 1506. This bill established an investigatory unit at the California Department of Justice for all officer-involved shootings regardless of justification. Transcripts published by the California Office of the Attorney General involving direct and indirect officer statements with voluntary, compelled, or prepared statements were analyzed. Drawing on abductive reasoning was used to determine compatibility of existing theory with emergent themes from the available transcripts. From first-pass human coding, sensemaking theory was identified as the most compatible theory for analysis. Usage of multi-agent large language models (LLMs) supplemented coding and theme identification and provided a quantitative sentiment analysis through sensemaking theory. This study ($n = 27$) found the formation of a mental picture, desire to stop a threat, and own identity formations were key themes that shaped an officer's use of force. These key themes aligned with Weick's sensemaking theory as plausibility over accuracy among situational cues, enacting environments, and individual and social identity relations. A supplemental quantitative analysis through LLMs found, on average, use of force was justified ($\bar{x} = 6.9$) and unavoidable ($\bar{x} = 4.5$), but that sensemaking was partially incomplete ($\bar{x} = 6.4$). This study offers key insights into how officers conceptualize use of force and provides an underused methodological lens in conceptualizing use-of-force.

Introduction

Law enforcement officer's (LEO) use-of-force in the United States has continuously been in the public eye and a political point of contention. The national struggle is infamously brought to national light from the beating of Rodney King in 1992 which was one of the first recorded incidents of police brutality by a bystander (Rothman, 2016). The efforts to reign in police was flung into national attention again after the murder of George Floyd by a Minneapolis police officer, which instigated the summer protests of *Black Lives Matter*. These incidents have larger societal implications, including loss of public institutional trust, cost to taxpayers, and extension of harm beyond the harmed resident to families and communities.

Use-of-force has traditionally been held to a legal definition and interpretation. The landmark Supreme Court, *Graham v. Connor* created the "objective reasonableness" standard, which is defined as what any reasonable officer would do, with regard to force, upon entering a situation not in hindsight. However, this leaves the ambiguity for what constitutes a reasonable officer. The Department of Justice has typically defined a reasonable officer's action through the use-of-force continuum that matches an officer's perception of risk with an appropriate response (National Institute of Justice, 2009). Some states have attempted to legislate more narrow definitions. In California, AB392 defines that officers shall only use lethal force only when necessary, after the death of Stephon Clark from a Sacramento Police Department officer. The incident, leading to public outcry, led to a settlement of \$1.7 million with the Clark family. Without a defined standard, LEO's conceptualization, reasoning, and justification for their use of force may be based on the provided training and protocols of their respective agency along with the "totality of circumstances (Goldman, 2021)."

Academic insight into use-of-force have provided varied perspectives. A systematic

review of 53 articles found the majority of articles could not cite a conceptual definition nor a consistent definition in defining use of force. (Frank Klahm IV et al., 2014). Within the legal realm and public safety, scholars have found that legal standards have left piecemeal ambiguity for what an officer is able to do (Goldman, 2021; Pham, 2022). Among greater social research, public perception and police officer perception in what constitutes use-of-force has been in conflict and in contrast with each other (Rojek et al., 2012). Public expectations of the use of force has been found to be in conflict with the legal expectations and definitions (Mourtgos & Adams, 2020). Ultimately, these conflicts arise through the central thesis provided by Arévalo in that policing is inherently made dangerous through the imperative to respond to danger and the need to protect oneself from danger (Sierra-Arévalo, 2021).

With mounting pressure after George Floyd, the California legislature passed AB 1506, which mandated a state review of all use-of-force incidents involving shootings. California is estimated to experience 195 fatal use of force incidents and is ranked first in fatal officer shootings among large U.S. states (0.38 fatal shootings per 100,000) (Hosseini, 2020). Investigators with the California Office of the Attorney General (OAG) are charged with interviewing and assessing the entire circumstance of a shooting (Office of the Attorney General, California Department of Justice, 2021). We intend to use these reports to develop a greater understanding of how officers define use of force.

Study Objectives

This proposed study will evaluate available officer statements, interviews, and transcripts issued in published AB 1506 reports to explore the conceptualization of force by an involved officer. Rather than applying a use-of-force definition to the officers, this study instead assumes that officers have an inherent view of applying use-of-force and seeks to illuminate their inherent

definition. Thus, the purpose of this study is to identify themes within officer interviews and to characterize the sensemaking process of LEOs.

Methods

Data Collection

We evaluated the case archive of AB 1506 reports California Attorney General's Office pertaining to involved officer statements were collected using the PyPDF package. Officer testimony is provided by either voluntary or compelled interviews or by written statements. Testimony is collected by investigators at the California Department of Justice, California Highway Patrol, and the respective police agencies. For a report to be included, an officer statement or interview from either the involved or responding officer had to be mentioned in the report. Interviews were not standardized, as officers were allowed to review body worn cameras during their interviews.

Analysis

First pass coding and initial thematic development will be completed through human coders using Taguette (Taguette Project 1.4.1) to develop relevant codes for the project. An abductive approach was developed where myself, as the qualitative instrument (Wa-Mbaleka, 2020), developed an inductive coding for analysis and allowed themes to emerge from the data consistent with a qualitative interpretive approach. Through several transcripts and iterative review, I centered on a deductive framework using Karl Weick's sensemaking theory as the basis for analysis (Weick, 1995).

Using this theory, I pursued natural language processing (NLP) through a large language model (LLM) using a multi-agent qualitative, thematic analysis grounded by the sensemaking process (Rosa et al., 2021) of LEOs during a use-of-force incident. LLMs were instructed to

follow the provided coding manual and to develop new codes as needed. Through the development of these codes, the LLMs were instructed to use sensemaking theory to quantitatively code the transcripts on a scale from 1 to 10, the likelihood of necessary use of force, the avoidability of a fatal encounter, and the completeness of sensemaking. Each LLM was asked to self-define these items and provide justification for their scoring. The final output of the LLMs consisted of the following: a list of codes per transcript, a quantitatively scored transcript, and identification of themes across the analysis of officer-involved shootings.

Positionality and Reflexivity

My positionality – I am a graduate student that brings my Filipino community into research and the learned public health lens grounded by prevention of morbidity and mortality. I take a critical approach informed by an ethnic studies and liberation lens who found myself organizing against police brutality after a community member died at the hands of police during a mental health crisis. At the same time, I have been faced with the challenges of policing as I was within the scene of the senseless murder of a police officer by a random gunman. Thus, I recognize the danger of policing, when it is needed, but also when people are placed in danger because of the system that resorts to policing as the only option. To mitigate against my preconceived notions and bias, I implemented reflection and journaling after each transcript. I documented analytic and reflexive memos to enhance the rigor of the study and ultimately provides a contribution in reducing harm caused by a legal intervention.

Generative AI Acknowledgement. A multi-agent programming workflow was developed to create a web-scraper flow Python 3.15 with ChatGPT GPT-5.2 (OpenAI, December 2056 build) and Claude Sonnet 4.5 (Anthropic, September 2025 build). The development of a multi-agent large language model for thematic and NLP analysis used Claude Sonnet 4.6

(Anthropic, March 2026 build) and ChatGPT GPT 5.4 Deep Thinking (OpenAI, February 2026 build). Originally, Gemini for Education Gemini 3.0 (Google, December 2025 build) was used for the multi-agent NLP workflow but was found to be hallucinating cases and descriptions.

Results

Qualitative Analysis

Among the 33 published reports, 27 transcripts were available for analysis. Excluded transcripts did not have any provided interviews ($n = 3$) or had data extraction errors that were not resolved in the computational workflow ($n = 3$). Through a combination of human and computational coding (Table 1), three central themes were identified: an officer's need to form a mental picture; an officer mentality of needing to stop the danger; and officer own- and social-identity formation that takes a part in their situational analysis (Figure 1). Examples of quoted lines with these themes are provided in Table 2. Consistent with sensemaking theory, these themes were translated to (1) plausibility versus accuracy through the extraction of cues, (2) officers repeatedly enacting simulated environments, and (3) individual and social identity.

Formation of a Mental Picture. Officer formation of their mental picture was centered on the idea that a threatening situation was occurring. As a result, officers tended to evaluate a situation based on two ideas: potential threat cues and potential for collateral damage. In one instance, one officer cited:

“Based on [decedent’s] baggy clothing and mannerisms, as well as W-3’s report that [decedent] had gestured at his waistband in a threatening manner, Officer Castaneda felt that frisking [decedent] was necessary” while also evaluating the whole of the situation, later stating, *“people began coming out from the surrounding businesses, and Officer Castaneda was trying to make sure that no one was down range.”*

This constant evaluation and re-evaluation of their mental picture establishes that officers are looking for the plausibility of a threat based on the acquired cues and pre-provided knowledge of a situation. Weick's sensemaking theory defines this as plausibility over accuracy, in which an officer tends to be processing incomplete information, but rather the likelihood of a threat.

Stopping the Danger. At the same time, there is a reiteration of danger and the need to stop the danger or threat. Officers defined this idea in three ways: identifying lives in danger, rationalizing that no other option existed for the officer, and the shared centralization that LEOs are sent to neutralize a threat.

Officers would rationalize their escalation of force through their rationalization that someone's life was in danger, whether their own, their colleagues or others around the scene. As one officer stated: *"he was going to be able to close that gap again because we still so close [sic], he's still moving. He's not stopping. He said he's going to fuck me up. He's going to kill me."*

Within this escalation was officers recognizing that they had run out of options for the officer as shown by direct quotes that said "Hey, I'm out of options. I've given this guy every opportunity. I've tried to deescalate as many times as I could, but he he's dead set on trying to take my gun from me." To an extreme, some reluctantly came to that conclusion, stating "Please don't reach for the gun. Please don't reach for the gun. I don't want to have to make a decision." While not all officers identified this language, this was a pervasive scripting that led to the rationalization of force for an officer.

Lastly, officers approached situations, whether in planned scenarios or in response to an incident with the idea that a threat needed to be stopped. This was a consistent narrative

regardless of the situation. For one officer who was leading a tactical response team, he chose not to intervene with the team, stating: *"He could tell them to slow down so they would have better situational awareness but did not stop the team because of the possibility of ongoing assaults."* In situations with a set response, the goal was apprehension and officers needed to get the person: *"100 percent, that's our guy."* Around this same time, Rodriguez stated, *"That's our guy. That's our guy."* Detective Chalmers responded, *"All right. Let's go and take him down."*

This key thematic domain is central to the Weick's idea of enacted environments that shape a person's behavior. Officers shape their idea of a threat environment, which informs their response to a situation. And for these responding officers, the environment was threatening and needed to be controlled.

Identity Formation. Lastly, officers bring an identity that relates to themselves, the people around them, and how they define themselves, meaning the officers fulfill an administrative, power, or personal role. In an administrative role, one officer spoke of being worried for their trainee during an attempted apprehension. In a power role, one officer by being taller amongst other officers would be leading a group responding to a mass casualty incident. And in a personal role, one officer mentioned that they thought about their child as it related to shooting at a decedent. These roles establish how an officer responds, reacts, and reflects upon a situation.

Part of the identity formation within a situation is the experience of the responding LEO, such as looking for deviations from typical patterns or learned experience that reinforce their decision making. For complex and changing situations, this information is often what an officer relies upon in one example:

"Officer Van Dragt indicated he has significant experience with traffic collisions

involving pedestrians and that there are usually injuries, and very often, fatalities. "It was just very confusing to me because this pedestrian had supposedly been run over by – um – cars, and then now, all of a sudden, there was a pedestrian that's standing on the shoulder and running at me, like, so quick. It just – it just – it just – very – happened very quick."

In other cases, deviations from an established situation primed the response of the officers:

Officer Meadows found [decedent]'s response strange since the typical response when officers approached in this way was either worry, a sense of urgency, or compliance. But [decedent] seemed nonchalant and in Officer Meadows' experience, people like that were unpredictable.

However, officers often stated a reinforcement of their own experience to gauge their response:

"In his experience, people fleeing a location meant they were in fear for their lives."

These ideas are central to Weick's sensemaking concept of Individual and Social Identity, in which a person brings into a situation a relationship with themselves and others to contribute to a collective sensemaking for all.

Quantitative Analysis

Given the development of these three themes, we prompted the LLMs to provide a sentiment scoring for the following three themes: (1) the necessity of using force, (2) the avoidability of a fatal encounter, and (3) completeness of the sensemaking with all provided cues and the quality of cues. Both Sonnet 4.6 and GPT 5.4 scored similarly for avoidability (4.3 and 4.6, respectively; $\bar{x} = 4.5$) but greatly differed in justifying use of force (7.4 and 6.4, respectively; $\bar{x} = 6.9$) and sensemaking completeness (6.7 and 5.6, respectively; $\bar{x} = 6.4$). Generally, this means the LLMs tended to agree that use of force was justified but not necessarily avoidable. At

the same time, sensemaking cues and factors were moderately available in each described situation.

Discussion

This study represents the first use of a paired human- and computational- approach in applying sensemaking to the study of use-of-force among LEOs. The use of sensemaking provides a powerful analytical lens in evaluating the actions of LEOs. Emerging from the transcripts were different iterations of threat analysis, in which LEO training, experience, and identity reinforce their visualization of a threat, need to neutralize a threat, and responses to a threat. The underlying concept of a threat means officers make sense of their situation by the plausibility of a threat, enacting and acting upon an environment that is threatening, and the concept of a threat is being reinforced by their identity and others around them. Together, these themes reinforce an idea that use-of-force extends beyond a formal definition but a collective idea that has been informally institutionalized by policing institutions (Waegel, 1984).

A shared theme identified by the LLMs but not considered in this analysis was the idea of police tactics as being instigators of a threatening situation, which is reinforced by officer training to command and control a situation (Wood et al., 2020) along with the need to make decisions in seconds (Zimmerman, 2022). Experience also likely plays a role in use of force as a systematized review found officers more tactically trained officers are able to reason and identify the necessity in using force more readily (Jenkins et al., 2024).

LLM Scoring Disparity. Sonnet 4.6 and GPT-5.4 differed in the quantitative scoring. In reviewing the provided LLM, GPT-5.4 held a more conservative stance on use-of-force justifiability as ambiguity of situations and considerations for alternative escalation of force – essentially invoking a hindsight approach. In addition, GPT-5.4 scored sensemaking lower as it

correctly identified several interviews did not have the direct statement or interview from the shooting officer. However, GPT 5.4 applied this lack of information to the scoring even after several iterations. The scoring differences also likely reflect the ambiguity of use of force with its inconsistent theory and unstandardized data collection (McLean et al., 2022).

Limitations. This analysis is subject to three core limitations. First, interviews are not conducted in a consistent manner. Second, not all interviews had the involved officer thus complicating the record of use of force. Lastly, we treated situations the same rather than separating by the type of response for a situation. However, given these limitations, the purpose of this analysis was to establish how officers conceptualize use of force regardless of whether they are involved despite the incomplete record.

Conclusions

Overall, these findings reinforce a formal and informal organizational sensemaking process among California LEOs. The sensemaking process that is reiterated through these reports reinforce that officer safety, department policy, and LEO culture are undercurrents are primary drivers in the use of force (Gau & Brunson, 2010; Sierra-Arévalo, 2021; Stoughton, 2014). We find that LEOs are primed by training and institutional and social reinforcement to respond and neutralize a threat. These findings could ground an idea of necessary use of force while also providing key training areas to change LEO behavior.

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Table 1: Initial Thematic Groupings

Claude	Gemini	Self
1) Plausibility versus accuracy 1) Most possible threat scenario 2) Threat architecture 1) Framing of incident and mental model prior to arrival 3) Compression of time 1) Force escalation and training 4) Incomplete record 5) Perceptual degradation 6) Positioning 7) Mental Crises 8) Exhausted Options	1) Cue compression and threat amplification 2) Retrospective justification over real time deliberation 3) Temporal compression ("split-second" decision making") 4) Institutionally constrained narratives 5) Plausibility over accuracy 6) Normalization of uncertainty and ambiguity 7) Tension between necessity and avoidability	1) Threat to self and others. 1) Once an officer establishes that the person may pose a threat to one's self, responding officers or others, they are pre-empted to use force 2) Actively looking for a threat 2) No other option 3) Based on training 1) Attending to conflict 2) What to look for

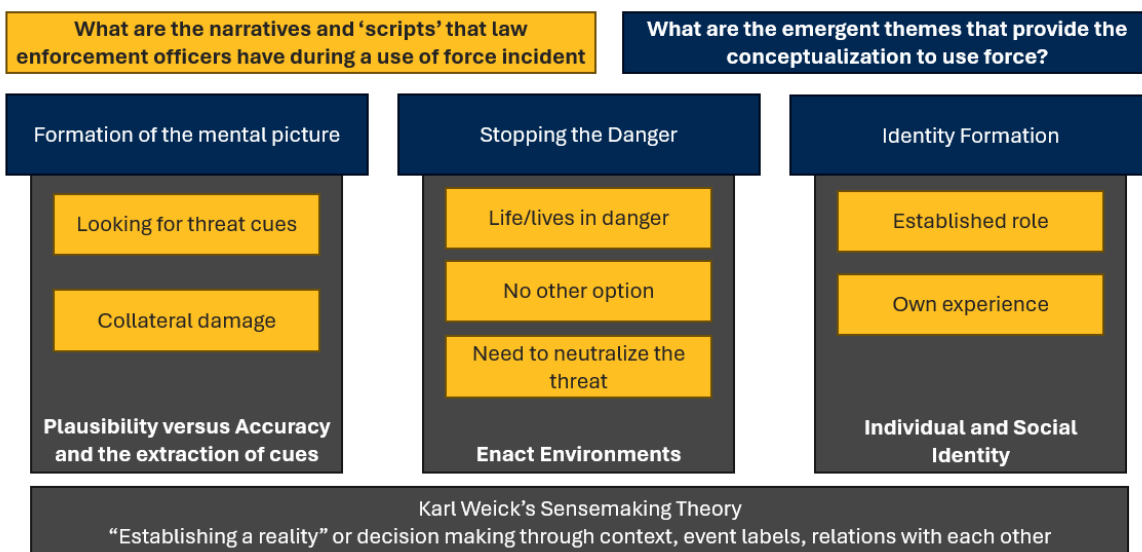
Table 2: Matrix of quotes from the analysis in relation to themes and sub-themes

Formation of a Mental Picture	Looking for Threat Cues	"For what?" Before Officer Orozco could advise [decedent] he was under arrest, [decedent] placed his right hand into his sweater pocket in an "aggressive" manner.
	Collateral Damage	"I thought I was gonna take one...I'm just gonna probably take one, hopefully it misses but I can't shoot right now, because it looks like people are home. "
Stopping the danger	Life in Danger	he was going to be able to close that gap again because we still so close [sic], he's still moving. He's not stopping. He said he's going to fuck me up. He's going to kill me
	No Other Option	"Please don't reach for the gun. Please don't reach for the gun. I don't want to have to make a decision." He thought that [decedent] was going to pull out his gun, shoot either of the officers, and kill them.

	Need to Neutralize a Threat	He could tell them to slow down so they would have better situational awareness but did not stop the team because of the possibility of ongoing assaults .
Identity Formation	Established Role	I'm worried for myself, I'm worried for my trainee , I'm worried for anybody who's – might look at this guy wrong if he ends up getting ahold of us.
	Experience	reinforced Mazur's belief that a crime was in progress inside the store: In his experience , people fleeing a location meant they were in fear for their lives.

Table 3: Average quantitative sentiment analysis of the transcripts (n = 27)

LLM	Force Necessity	Avoidability of a Fatal Encounter	Completeness of Sensemaking
Sonnet 4.6 (Claude)	7.4	4.3	6.7
GPT 5.4 (OpenAI)	6.4	4.6	6.0

Figure 1: Thematic map of emergent themes and overlap with sensemaking theory

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